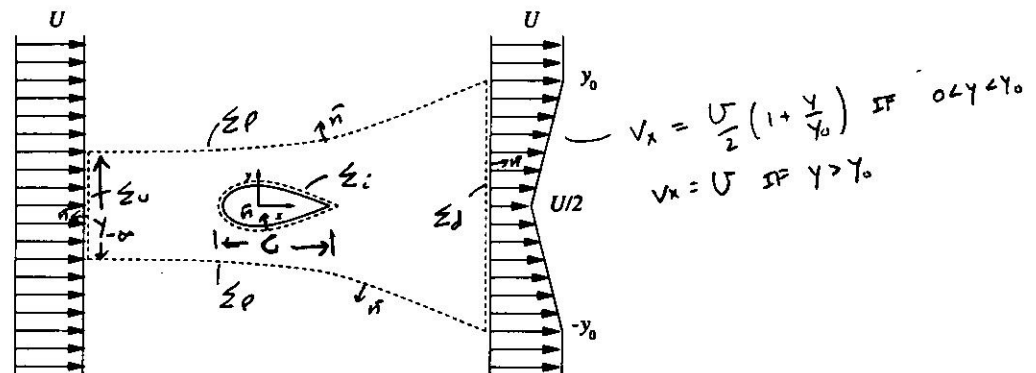


The experimental tests of an airfoil in a wind tunnel indicate that the velocity profile upstream remains uniform, whereas behind the airfoil there exists a wake region with significant velocity deficit whose thickness increases downstream, resulting in the flow field sketched in the figure below. For the airfoil tested, the velocity measurements at a given downstream location reveal that the velocity in the wake varies linearly from the value $u = U/2$ at the centerline to reach the value $u = U$ at the edge of the wake, located at $y = y_0$. To analyze the problem, consider a control volume limited laterally by the streamlines that extend from $x \rightarrow -\infty$, $y = \pm y_{-\infty}$ ($y_{-\infty} < y_0$) to $x = x_0$, $y = \pm y_0$ and internally by the surface of the airfoil. Assuming that the air behaves as an incompressible fluid with density ρ and that viscous stresses are negligible both in the wake and along the bounding streamlines, obtain the value of $y_{-\infty}$ as a function of y_0 , as well as the value of the drag on the airfoil as a function of ρ , y_0 and U . In the analysis, assume that on the outer boundary of the control volume the pressure differences from the ambient value p_{∞} are negligible, and so is the effect of gravity on the air flow.



STEADY
 ~~$\frac{d}{dt} \int_{V_c} \rho dV$~~ + ~~$\int_{\Sigma_c} \rho (\vec{v} - \vec{v}_c) \cdot \vec{n} d\sigma = 0$~~ ⇒ $\int_{\Sigma_c} \vec{v} \cdot \vec{n} d\sigma = 0$ (BECAUSE $\vec{v} \cdot \vec{n} = 0$ (NO FLUID CROSSES Σ_P AND Σ_I))

FIXED CONTROL VOLUME

$\int_{\Sigma_U} \vec{v} \cdot \vec{n} d\sigma = -U \int d\sigma = -2y_{-\infty} U$
 $\int_{\Sigma_D} \vec{v} \cdot \vec{n} d\sigma = 2 \int_0^{y_0} v_x dy = 2 \int_0^{y_0} \frac{U}{2} \left(1 + \frac{y}{y_0}\right) dy = 2 \cdot \frac{U}{2} \cdot \frac{3}{2} y_0 = 3U y_0$

FINALLY, $-2y_{-\infty} U + 3U y_0 = 0 \Rightarrow \boxed{y_{-\infty} = \frac{3}{4} y_0}$

THE FORCE IS GIVEN BY

$\vec{F}_{a \rightarrow \text{airfoil}} = - \int_{\Sigma_a} p \vec{n} d\sigma + \int_{\Sigma_a} \vec{\tau} \cdot \vec{n} d\sigma$

MOMENTUM CONSERVATION APPLIED TO THE CONTROL VOLUME GIVES

~~$\frac{d}{dt} \int_{V_c} \rho \vec{v} dV$~~ + ~~$\int_{\Sigma_c} \rho \vec{v} [(\vec{v} - \vec{v}_c) \cdot \vec{n}] d\sigma$~~ = ~~$\int_{V_c} \rho \vec{g} dV$~~ - $\int_{\Sigma_c} p \vec{n} d\sigma + \int_{\Sigma_c} \vec{\tau} \cdot \vec{n} d\sigma$ (NO GRAVITY)

$\int_{\Sigma_U + \Sigma_D + \Sigma_P + \Sigma_I} \rho \vec{v} \cdot \vec{n} d\sigma = -2\rho U^2 y_{-\infty} \vec{e}_x + 2\rho \left(\frac{U}{2}\right)^2 \vec{e}_x \int \left(1 + \frac{y}{y_0}\right)^2 dy = -\frac{1}{3} \rho U^2 y_0 \vec{e}_x$

$-\int_{\Sigma_U + \Sigma_D + \Sigma_P + \Sigma_I} p \vec{n} d\sigma \approx -\int_{\Sigma_U + \Sigma_D + \Sigma_P} p_a \vec{n} d\sigma - \int_{\Sigma_I} p \vec{n} d\sigma, \quad \int_{\Sigma_U + \Sigma_D + \Sigma_P + \Sigma_I} \vec{\tau} \cdot \vec{n} d\sigma \approx \int_{\Sigma_I} \vec{\tau} \cdot \vec{n} d\sigma$ (BECAUSE $\frac{U y_0}{\mu} \gg 1$)

$-\frac{1}{3} \rho U^2 y_0 \vec{e}_x \approx -\int_{\Sigma_I} p \vec{n} d\sigma + \int_{\Sigma_I} \vec{\tau} \cdot \vec{n} d\sigma = -\vec{F}_{a \rightarrow \text{airfoil}} \Rightarrow \boxed{\vec{F}_{a \rightarrow \text{airfoil}} = \frac{1}{3} \rho U^2 y_0 \vec{e}_x}$